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Optimizing agricultural water use: Crop water requirement and irrigation scheduling of tomato & soybean using FAO CropWat 8.0 model in Chandrapur district, Maharashtra, India

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ABSTRACT

Effective water management is essential for maximizing crop production, water efficiency, and the health of crops, especially in semi-arid regions. This study assesses the crop water requirements and irrigation scheduling for tomato and soybean crops in Chandrapur, Maharashtra, using the FAO CropWat 8.0 model. A combination of ET_n estimations, effective rainfall, irrigation water requirement, crop coefficient (Kc), crop evapotranspiration (ETc), and soil moisture monitoring was done to establish the water requirements and irrigation scheduling for the key crops. The results show that tomatoes will require an adequate water supply during the flowering and fruiting period. The crop water requirement is 434.3 mm/dec and an irrigation requirement of 342 mm/dec for tomatoes. The soybean crop requires water about 309.3mm/dec and 7.6mm/dec for irrigation, only during the initial and late periods. On the other hand, the CropWat model projected a water need of 35.3 mm net irrigation and 50.4 mm gross irrigation at 70 % field efficiency for the thrive production of Soybean. For tomato, the model estimated a water requirement of 365.7 mm net irrigation and 522.5 mm gross irrigation at 70 % field efficiency for best production. The study emphasizes the need to align irrigation management with ET, soil moisture, and climatic conditions for sustainable agriculture. Future initiatives involving remote sensing, AI-induced irrigation, and climate-adaptive strategies can boost water conservation. The findings can assist researchers, policymakers, and farmers in ensuring more effective irrigation management in water-scarce regions.

1. Introduction

In the context of global warming and climate change, the interest of researchers and policymakers has been deeply aroused, especially in the field of agriculture and hydrology [1,2]. According to the UN World Water Development Report, globally agriculture is the largest consumer of freshwater, accounting for roughly 70 % of all freshwater withdrawals, while over 60 % of the irrigated water goes to waste, raising red flags on the suitability of the present-day irrigation management methods [3,4]. Approximately 84 % of India's total available water resources are used for irrigation, yet a substantial amount is lost due to inefficient water management practices [5]. Ensuring adequate water availability is vital for stable crop production and must be prioritized to address the increasing demand for food and nutritional security driven by global population growth [6]. This is particularly necessary for vegetables, to fulfill the dietary needs of the growing world population in the present time as well as the future [7]. Among these, tomato and soybean are acclaimed eminence as highly consumed and most nutritionally endowed vegetable crops cultivated widely across different regions of the world [8]. Normal tomato cultivation demands enormous quantities of irrigation water in the range of 600–800 mm, depending on the environment [9]. Major irrigation management is always paramount to tomato production, because inadequate irrigation can easily impair the quality and yield of tomatoes due to the high sensitivity of water conditions [10]. In the scenario of climate change and global water scarcity, soybean cultivation played an important role in rain-fed areas for agricultural production [11,12]. Planning and time series optimization of agro climatic parameters such as crop evapotranspiration (ET), crop water requirement (CWR), net irrigation requirement (NIR), and water sensitivity (WS) are very essential to crop water management, like soybean [13,14]. Farmer's shortsighted irrigation practices not only reduce crop production and quality but also aggravate the squandering of freshwater resources, and as a result, the water scarcity increases around the globe [15]. Understanding the intricate interactions between climate, water, and crop growth needs to be a priority area in the entire world. Since water is the most crucial input for crop production, therefore accu-

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rate planning of this precious resource is necessary. The water requirement of crops varies widely from crop to crop, and during the entire crop growth period of an individual crop. In other words, knowledge of crop water requirements is necessary for effective irrigation scheduling, water balance, canal design capacities, regional drainage, and water resources planning. Efficient irrigation scheduling requires an accurate estimation of crop evapotranspiration [16]. Moreover, the determination of the net irrigation requirement is necessary, so it can be properly designed for the irrigation system and the establishment of irrigation schedules. CropWat 8.0 is an FAO-designated simulation model to calculate CWR and irrigation needs based on soil, climate, and crop data [17]. All over the world, researchers [17–23] have utilized the CropWat model to investigate CWR and irrigation scheduling for a diverse range of crops with great results. Importantly, this research intersects with two United Nations Sustainable Development Goals (SDGs). SDG 2 calls for the elimination of hunger through improved agricultural productivity and sustainable food systems [24], while SDG 13 urges nations to take urgent action to combat climate change and its impacts on agricultural food production [25]. Gabr (2023) [26] assesses the impact of climate change on future irrigation water requirements in Upper Egypt using the CropWat model and climate projection scenarios. Due to climate change, it finds a significant increase in water demand for some crops in water-scarce areas. Complementing this, Gabr et al. (2024) [27] demonstrated through CropWat 8 and WaterGems simulations that suggest evapotranspiration in Egypt's western desert will increase by over 12 % by 2100 under RCP 8.5, further stressing the limited water supply. Highlighting the urgency for adaptive water resource management to sustain agricultural productivity and eliminate hunger through food security in water-scarce environments. Chandrapur district, situated in the semi-arid region of Maharashtra, experiences a hot and dry climate with average annual rainfall around 1249.4 mm, predominantly during the monsoon months of June to September. The district's agriculture is heavily reliant on monsoon rains, making it vulnerable to climatic variability and water scarcity. In recent years, the region has witnessed challenges such as erratic rainfall patterns with drought and flood-like conditions, affecting crop yields and water availability. Conversely, the 2020 Vidarbha floods affected over 92,000 people and a large number of agricultural lands [28], including Chandrapur district, highlighting the region's vulnerability to extreme weather events [29]. Tomato and soybean are among the key crops cultivated in Chandrapur. However, soybean cultivation has faced setbacks due to outbreaks of yellow mosaic disease, affecting approximately 67,766 hectares and leading to significant yield losses [30,31]. These challenges underscore the need for efficient water management practices to ensure sustainable agricultural production. This study is the first to assess the crop water requirement (CWR) and irrigation scheduling for tomato and soybean in Chandrapur district of Maharashtra. No prior research has applied this model to these specific crops in this geographic context. Additionally, it offers practical irrigation calendars and net irrigation demand estimates for tomato and soybean. Several studies in South India have explored water requirements for tomato and soybean, but this study takes a distinct approach in terms of methodology and scope. Shrikant et al. (2023) [32] used a lysimeter to estimate tomato evapotranspiration in Karnataka, while Guhan et al. (2024) [33] applied the AquaCrop model to assess climate impacts on tomato and water needs in Tamil Nadu. Studies like Kashyapi and Bahot (2012) [34] have focused on soybean water needs, and none have integrated FAO tools for irrigation planning. Phad et al. (2024) [35] focused on the estimation of crop water requirement on soybeans in the Marathwada region; their study was only focused on crop water requirement (CWR), without incorporating irrigation scheduling frameworks. On the other hand, Hemareddy et al. (2025) [36] applied the CropWat model for soybean under climate change scenarios in north interior Karnataka; they primarily emphasized climate variability impacts rather than a full irrigation scheduling strategy or multi-crop water optimization. Therefore, the objective of our study is to calculate the crop water requirement and irrigation schedule of tomato and soybean crops. We employed the CropWat 8.0 model for crop water requirement, which is a widely recognized tool for water resource management in agriculture. The study utilized the FAO Penman-Monteith method to calculate the reference evapotranspiration (ETo). Additionally, we assessed various factors, including irrigation water requirements (IWR), effective rainfall, and cultivation time for major crops. Also, Key soil parameters, crop coefficient (Kc), and crop evapotranspiration (ETc) were considered to calculate crop water requirement & irrigation scheduling in Chandrapur district of Maharashtra, India. We have done a proper systematic analysis for crop water requirement (CWR) & irrigation scheduling (IRS) of tomato and soybean (Fig. 3); therefore this study is very crucial and helpful to policymakers, researchers, and agricultural stakeholders to make decisions about sustainable water management for agriculture in the future.

Table 1
Climatic, Soil, Crop data and Parameters.

Data	Parameters
Climatic & Meteorological	Humidity and wind speed, Sunshine hours, Minimum and Maximum temperature, rainfall
Soil	Maximum rooting depth, Total available Soil Moisture, Maximum rain infiltration rate
Crop	Crop coefficient (Kc) values for different stages of crop

2. Study area

The study was conducted in Chandrapur, a district in the eastern part of Maharashtra, India, located 19°30′N to 20°45′N latitude and 78°46′E to 80°55′E longitude (Fig. 1). The semi-arid climate exists in an area of 11,443 km^2, receiving an average annual rainfall between 1200 to 1400 mm, mostly in the southwest monsoon. Summer periods may reach temperatures as high as 48 °C and drop to 10 °C in winter [37].

3. Materials and methods

Several datasets were gathered, such as wind speed, sunshine hours, rainfall, relative humidity, maximum and minimum temperature, and soil type from ClimWat 2.0 and CropWat 8.0 software (Table 1).

3.1. CropWat 8.0 model

CropWat is a decision-support system developed by Food and Agriculture Organization (FAO), whose main functions are to calculate reference evapotranspiration, crop water requirements, crop irrigation requirements, develop Irrigation schedules under various management conditions, and a scheme of water supply [38]. The Penman-Monteith method, developed by Smith (1998) [39], is used in the computation of reference crop evapotranspiration. The estimates of evapotranspiration for the reference crops are used in computing the crop water requirement and in irrigation scheduling. CropWat calculates the irrigation requirement (Eqs. (7) & 8) for any crop at various stages of development, during the growing season on a weekly or monthly basis or as part of a cropping pattern in an irrigated area [40]. This model requires the main input parameters like soils, crop growth, and meteorological data

3.2. FAO Penman-Monteith method for ET_o

The FAO Penman-Monteith approach for estimating ETo, The FAO Penman-Monteith (Eqs. (1) & (2)) is as follows [19,41,42]:

$$\lambda ET = \frac{\triangle (Rn - G) + Pa\ Cp\frac{(es-ea)}{ra}}{\triangle + \gamma\left(1 + \frac{rs}{ra}\right)} \quad (1)$$

Where Rn is the net radiation,

G is the soil heat flux,

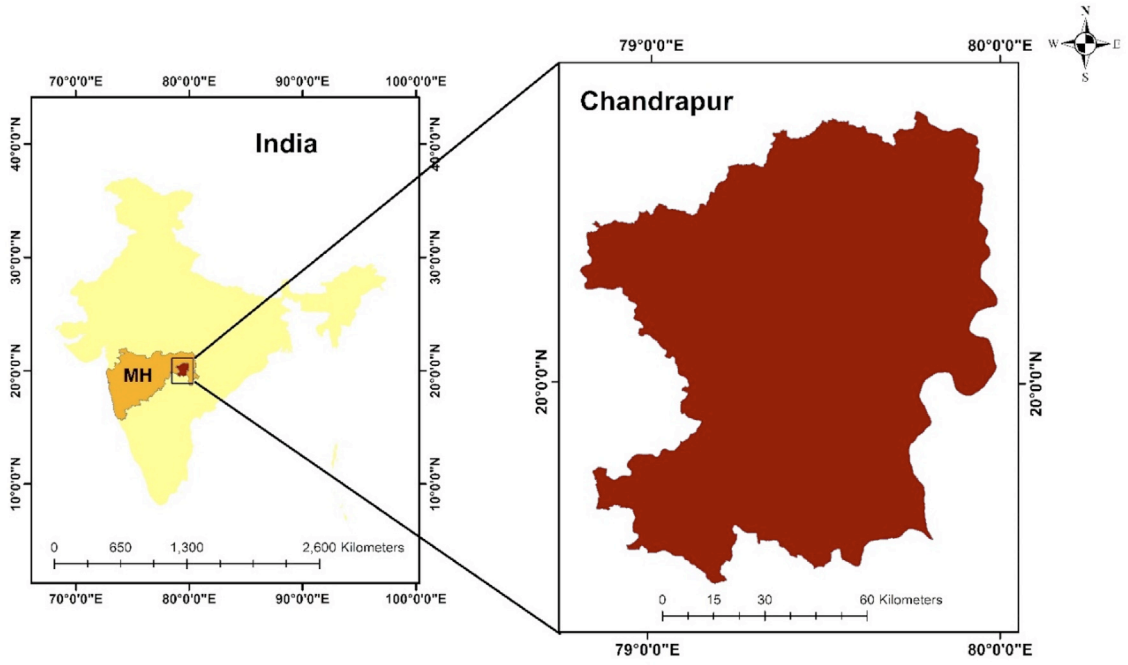


Fig. 1. Study area map, source authors.

(es − ea) is the vapor pressure deficit of the air,
Pa is the mean of air density at constant pressure,
Cp is the specific heat of the air,
Δ is the slope of the relationship between saturation vapor pressure and air temperature,
γ is a psychometric constant,
rs and ra are the surface and aerodynamic resistances, respectively

$$ET_O = \frac{0.408 \triangle \left(R_n - G\right) + \gamma \frac{900}{T+273} u_2 \left(e_s - e_a\right)}{\triangle + \gamma \left(1 + 0.34 u_2\right)} \quad (2)$$

Where, ETo = reference evapotranspiration [mm day-1]

Rn =net radiation at the crop surface [MJ m-2 day-1]
G = soil heat flux density [MJ m-2 day-1]
T =mean daily air temperature at 2 m height [°C]
u2 = wind speed at 2 m height [m s-1]
es = saturation vapor pressure [kPa]
ea = actual vapor pressure [kPa]
es – ea= saturation vapor pressure deficit [kPa]
Δ-the slope of the saturated water vapor pressure curve (kPa/c)

3.3. Irrigation water requirements (IWR)

CropWat can compute the daily water balance of the root zone with respect to root zone depletion at the end of the day [43]. To calculate the IWR (Eq. 3), the following equation is:

$$Dr,i = Dr,i - 1 - (p - ROi) - li - CRi + ETci + DPi \quad (3)$$

Where, Dr, i is the root zone depletion at the day's end i (mm), Dr,$i-1$ is the water content in the root zone at the previous day's end (mm), Pi is the precipitation on day i (mm), ROi is the surface soil runoff on day i (mm), Ii is the net irrigation depth on day i which infiltrates the soil (mm), CRi is the capillary rise from the groundwater table on day i (mm), ETci is the crop evapotranspiration on day i (mm), and DPi is the lost water of the root zone on day i (mm).

Table 2
Dates and months of crop showing and harvesting.

crop	Date of showing Date/Month	Date of Harvesting Date/Month
Tomato	1-Oct	3-Feb
Soybean	2-Jun	2-Sep

Source: National Academy of Agricultural Sciences, New Delhi, India, National Food Security Mission (NFSM).

3.4. Effective rainfall

Rainfall data were collected from the Chandrapur meteorological station using ClimWat 2.0. The effective rainfall was computed using the formulas (Eqs. (4) & (5)) of the USDA Soil Conservation Service (SCS) [41,21].

$$P_{eff} = \frac{P_{tot} \times \left(125 - 0.6 p_{tot}\right)}{125} \; for \; P_{tot} \leq \left(\frac{250}{3}\right) mm \quad (4)$$

$$P_{eff} = \frac{125}{3} + 0.1 \times \; P_{tot} \; for \; P_{tot} \geq \left(\frac{250}{3}\right) mm \quad (5)$$

Where,

P_{eff} = Effective rainfall (mm); P_{tot} = Total rainfall (mm)

3.5. Cultivation time of major crops & soil parameters

In our study, major cultivated crops such as tomato and soybean with the most suitable timetable for cultivation are below (Table 2) and soil parameters in Table 3:

3.6. Crop coefficient (Kc)

The crop coefficient (Kc) is the ratio of the crop ETc to the reference ETo and shows the assimilation of four principal factors such as crop height, crop-soil surface albedo, canopy resistance, and particularly soil evaporation in exposed soil [44]. For the crop evapotranspiration, we used standard FAO Kc values instead of local data because of the limited

Table 3
Key soil data and parameters, source: FAO & CropWat 8.0.

Soil Parameters	Name / Value	Unit	Description
Soil Name	Black Clay Soil	—	Soil type considered in the analysis
Total Available Soil Moisture (FC - WP)	200	mm/meter	The amount of moisture available to plants in the soil
Maximum Rain Infiltration Rate	30	mm/day	The maximum rate at which soil can absorb rainfall
Maximum Rooting Depth	900	centimeters	The maximum depth that roots can extend in the soil
Initial Soil Moisture Depletion (% of TAM)	50	%	Percentage of total available moisture depleted initially
Initial Available Soil Moisture	100	mm/meter	Soil moisture available at the start of the analysis

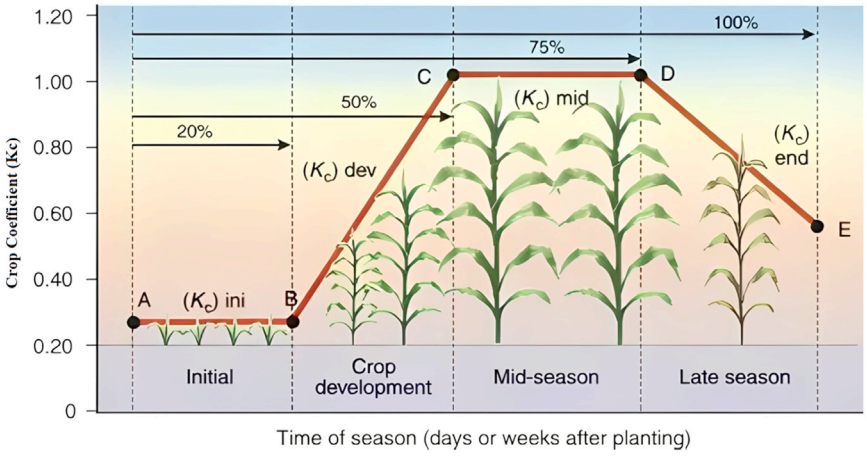


Fig. 2. Crop coefficient at various stages, source: Irmak, 2008 [64].

availability of region-specific crop coefficient data in Chandrapur. The crop coefficients for the growing season, considering several stages of crop development [44] (Fig. 2). Initial stage (first 20% of the season): Beginning from planting and continuing with plant emergence and germination. Crop development stage (20% to 50% of the season): The crop grows rapidly, and its leaf canopy expands. Consequently, the Kc value increases sharply as plant transpiration becomes the main form of water use. Mid-season stage (50% to 75% of the season): The crop reaches full canopy cover and its maximum size. The Kc value is at its peak and remains constant, representing the period of highest water demand. Late season stage (75% to 100% of the season): The crop begins to mature, and its leaves becomes senesce. The Kc value declines steadily, and the plant's water needs decrease leading up to harvest [45].

3.7. Crop evapotranspiration (ET_c)

The crop water requirements are the total amount of water that a crop will consume during the growing season. The Crop Coefficient Approach determines the ETc by orienting itself toward incorporating crop characteristics into the Crop coefficient and integration of the variable of different weather conditions into ETo. ET_c is calculated by using this formula [44] (Eq. (6)):

$$ET_C = KC \times ET_O \tag{6}$$

Where:

ETc= Evapotranspiration of crop
ETo= Reference crop evapotranspiration
Kc= Crop factor/Coefficient of the crop.

The net & gross irrigation required (Tables 6 & 8) was calculated by following (Eqs. VII & VIII):

$$NIR = ET_C - P_{eff} \tag{7}$$

Where, ET_c – Crop Evapotranspiration and P_{eff} - Effective rainfall (mm)

$$GIR = \frac{NIR}{e_a} \tag{8}$$

Where, e_a is irrigation operation performance, which is especially considered as the footprint of the adopted irrigation method. For surface irrigation, the considered value is 70 % [46].

3.8. Irrigation scheduling

Irrigation scheduling has long been employed to effectively control optimal water supply by raising inputs. Numerous indicative irrigation schedules can be generated and yield effects analyzed through the CropWat model for irrigation scheduling developed by FAO. Irrigation field programs should be designed based on yield reduction patterns and water use efficiency. These programs can be simulated under varying conditions, including water deficiency, rainfall events, and the application of supplemental irrigation [47].

The soil moisture deficit threshold serves as the basis for an irrigation schedule. This procedure calculates the limit of Readily Available Moisture (RAM) in the soil, when crops get stressed, see Eq. (9).

$$RAM = TAM \times F \tag{9}$$

Where TAM indicates total available moisture in the soil, measured as millimeters of water per meter in the soil horizon, see Eq. (10).

$$TAM = (Root\ depth) \times (Soil\ moisture\ capacity\ in\ mm\ or\ meters) \tag{10}$$

4. Results and discussion

4.1. Reference evapotranspiration(ETo)

The climatic parameters outlined for Chandrapur district demonstrate significant variations in temperature, humidity, wind speed, solar

Table 4
Calculation for ETo by Penman-Monteith method, source: CropWat.

Month	Min Temp (°C)	Max Temp (°C)	Humidity (%)	Wind (km/day)	Sun (hours)	Radiation (MJ/m^2/day)	ET_n (mm/day)
January	13.7	29.5	57	81	8.1	16.7	3.12
February	15.9	32.3	45	100	9	19.7	4.15
March	19.8	36.7	35	112	11.2	25	5.68
April	24.7	40	32	112	11.1	26.3	6.75
May	28.3	42.8	32	175	12.3	25	7.15
June	26.9	37.7	52	239	7.5	21.3	5.68
July	24.3	30.9	78	235	4.2	16.1	4.24
August	24.1	30.5	78	191	5.5	16.4	3.97
September	23.7	31.2	72	141	5.5	17	3.97
October	20.7	31.6	71	81	6.7	17.2	3.75
November	15.1	29.5	55	82	7.4	15.6	3.12
December	12.2	28.4	62	82	7.4	15.2	2.69

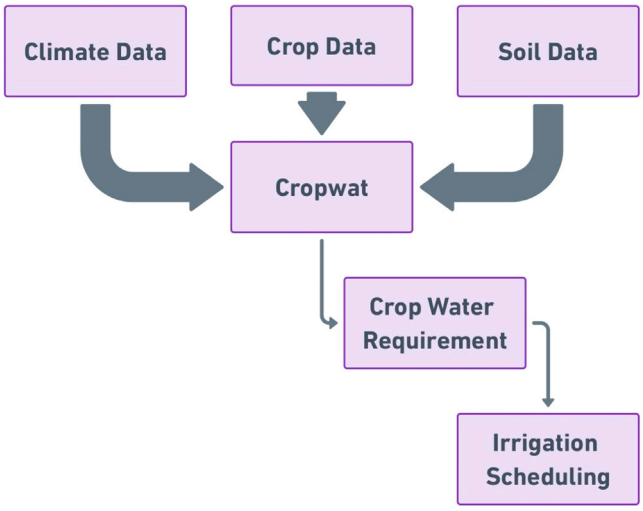


Fig. 3. Methodological flow chart of this study. source: author's.

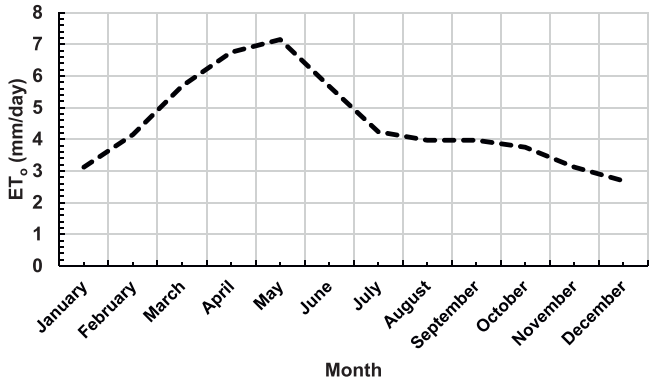


Fig. 4. Showing month-wise ETo source: CropWat.

radiation, and reference evapotranspiration (ET_n); all these have a direct bearing on the water requirements of tomato and soybean crops. This area witnesses an average ET_n of about 4.55 mm/day, with the highest values occurring in April (6.75 mm/day) and May (7.15 mm/day), due to higher temperature, lower humidity, and wind. High values ensure there are higher evaporative losses, which means the irrigation systems should be efficient, especially for the kharif and Rabi crops. Soybean, which is primarily cultivated in the kharif season (June-September). However, the peak ET_n values during May (7.15 mm/day) would imply that supplementary irrigation might be essential to avoid moisture stress, especially in the reproductive stages. Effective irrigation scheduling can help to mitigate the water shortage and encourage crop productivity. Tomato cultivated mainly in the Rabi seasons shows a higher sensitivity to water stress, especially during flowering and fruit-setting periods. In Rabi seasons (October-February), ET_n values fluctuate from 3.75 mm/day to 4.15 mm/day, showing moderate but constant water requirements. Due to low humidity and limited rainfall, this period necessitates frequent irrigation to sustain an adequate soil moisture level. In the kharif season, soybeans benefited from monsoon rain; however, the sporadic precipitation during the year supports extra irrigation intervention (Table 4) & (Fig. 4).

4.2. Rainfall & Effective rainfall

Fig. 5 shows the rainfall and effective rainfall of the Chandrapur district for each month. Total annual rainfall is 1346 mm, while effective rainfall is 754.9 mm using the USDA S.C. Method. Precipitation does not always directly contribute to crop water availability because of runoff, deep percolation, and evaporation losses. The maximum rainfall was recorded during the monsoon (June-September) months, with the highest being from July (404 mm) and August (340 mm). Effective rainfall is much less during the monsoon months (July & August), 165.4 mm respectively, suggesting massive losses of rainwater due to considerable runoff. Though rainfall is also considerable in June (181 mm) and September (234 mm), only 128.6 mm and 146.4 mm are effective for crop production. In contrast, the Rabi seasons (October-February) have minimal rainfall amounts. In October, we saw 72 mm of rainfall, which goes down to 5 mm in January. Effective rainfall is minimal, requiring additional irrigation for Rabi crops such as tomatoes. From November to April, precipitation is extremely limited, indicating the necessity for irrigation to maintain the growth of crops. The Kharif crop (soybean) benefits from the very high monsoonal rainfall. However, variations in effective rainfall suggest water conservation strategies to reduce runoff losses. For Rabi crops like tomatoes, where natural rainfall is insufficient, well-structured strategies are needed to meet crop water demands and sustain yield.

4.3. Attributes of tomato growth

Fig. 6 contains the basic irrigation and agronomy data on tomato cultivation. The crop coefficient (Kc) starts at 0.60 in the initial stage of development (30 days), rises to a maximum of 1.15 during mid-season (45 days), and then drops to 0.80 during the last stage (30 days), with a total growing season of 145 days. Rooting depth begins at 0.25 m and reaches up to a depth of 1.00 m during mid-season. Critical depletion ranges from 0.30 at the beginning to 0.50 at the end of the season. Mid-season yield response is highest in tomatoes (1.10), indicating maximum sensitivity to water stress. Crop height reaches 0.60 m in the mid-season.

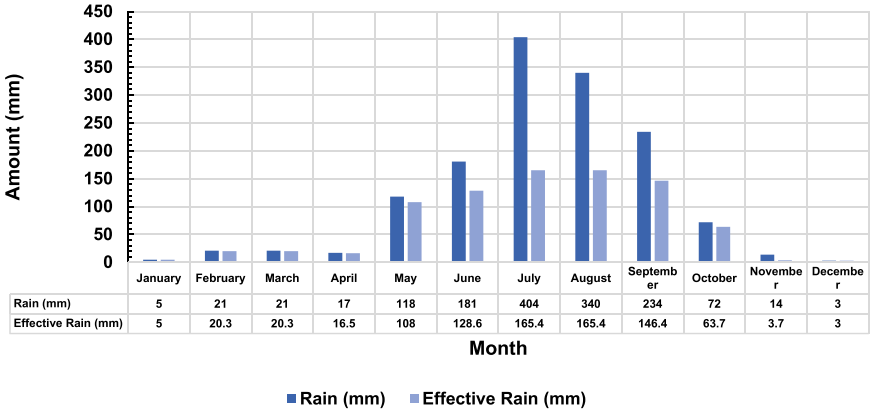


Fig. 5. Showing the rainfall and effective rainfall (mm) using the USDA S.C. method, source: CropWat.

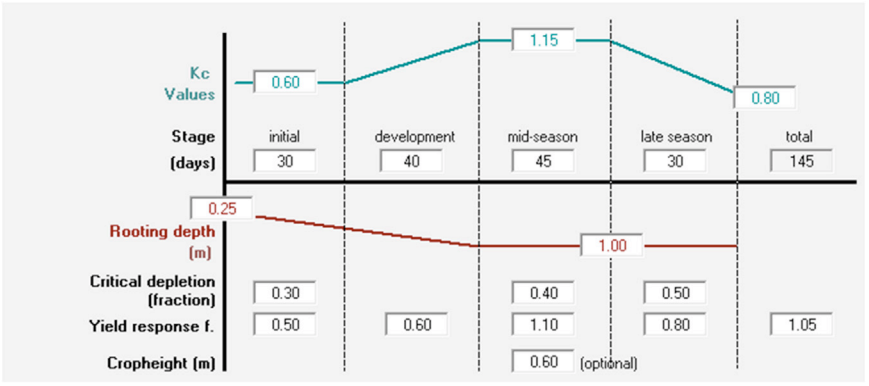


Fig. 6. Parameters of tomato growth with different stages in a season. source: CropWat.

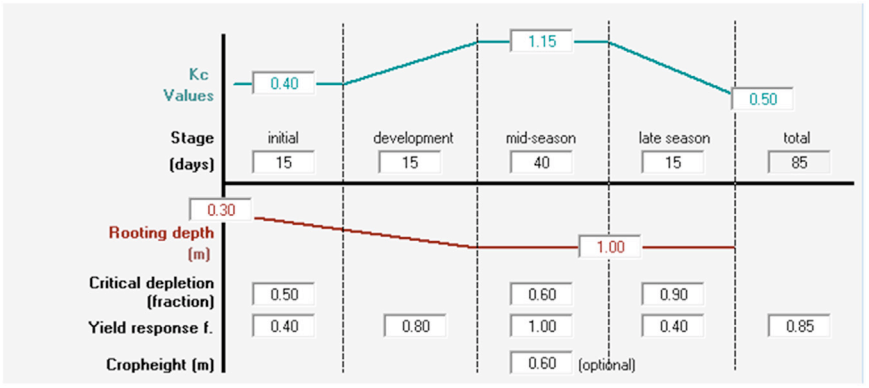


Fig. 7. Parameters of soybean growth with different stages in a season. source: CropWat.

Proper irrigation, especially during mid-season, ensures high-yield production.

4.4. Attributes of soybean growth

For soybeans (Fig. 7), the crop coefficient (Kc) increases from 0.40 in the beginning phase (15 days), attains a maximum of 1.15 during the mid-season phase (40 days), and then decreases to 0.50 in the late season (15 days) in a full cycle of 85 days. Rooting depth starts at about 0.30 m and ends at around 1.00 m in mid season, with critical depletion from 0.50 m in the early season, rising to a maximum of 0.90 m during the late season. The yield response factor is highest during the mid-season (1.00), indicating maximum sensitivity to water stress. The estimated crop height is 0.60 m during the mid-season. Satisfactory irrigation, especially in mid-season, plays a crucial role in ensuring proper yield.

4.5. Crop water requirement (ETc) and Irrigation Water Requirement (IWR) of tomato

Table 5 & Fig. 8 show how crop water requirements varied for tomato crops over various growth stages from October to February. The initial stage, which begins on October 1 - 2, has a crop coefficient (Kc) value of 0.6, indicating some low water demand. Irrigation requirements are minimal, from 0 to 2.8 mm/dec, with an effective rainfall being of great significance at the start of October. Additionally, during the development stage, starting on October 3 and ending on November 3, the crop coefficient gradually increases from 0.6 to 0.96, and evapotranspiration increases. This leads to relatively high irrigation needs, reaching 26.7 mm/dec in late November when rainfall becomes limited. The mid-season, beginning December 1 to January 2, suggests the highest water demands, with Kc ranging from 1.09 to 1.14. Daily values for ETc reach up to 3.56 mm/day, resulting in a high irrigation requirement of 28.6 mm and 34.5 mm/dec, due to limited rainfall. In the late stage,

Table 5
Month-wise crop water requirement and irrigation requirement of the tomato crop, source: CropWat.

Month	Decade	Stage	Kc	ETc (mm/day)	ETc (mm/dec)	Eff rain (mm/dec)	Irr. Req. (mm/dec)
Oct	1	Init	0.6	2.29	22.9	29.4	0
Oct	2	Init	0.6	2.25	22.5	19.7	2.8
Oct	3	Deve	0.6	2.14	23.5	14.7	8.9
Nov	1	Deve	0.69	2.31	23.1	8.9	14.3
Nov	2	Deve	0.82	2.61	26.1	2.7	23.4
Nov	3	Deve	0.96	2.89	28.9	2.1	26.7
Dec	1	Mid	1.09	3.04	30.4	1.8	28.6
Dec	2	Mid	1.14	2.95	29.5	0.4	29.1
Dec	3	Mid	1.14	3.15	34.7	0.8	33.8
Jan	1	Mid	1.14	3.39	33.9	1	32.9
Jan	2	Mid	1.14	3.56	35.6	1.1	34.5
Jan	3	Late	1.1	3.82	42	3	39
Feb	1	Late	0.99	3.75	37.5	5.5	32
Feb	2	Late	0.87	3.61	36.1	7.5	28.7
Feb	3	Late	0.8	3.74	7.5	1.8	7.5
total					434.3	100.5	342

Table 6
Stage-wise irrigation schedule of the tomato crop, source: CropWat.

Date	Day	Stage	Rain (mm)	Ks (fract.)	Eta (%)	Depl (%)	Net Irr (mm)	Deficit (mm)	Loss (mm)	Gr. Irr (mm)	Flow (l/s/ha)
1-Oct	1	Init	0	0.71	71	53	27.7	0	0	39.6	4.58
26-Oct	26	Init	0	1	100	31	33.1	0	0	47.3	0.22
18-Nov	49	Dev	0	1	100	36	55.8	0	0	79.8	0.4
13-Dec	74	Mid	0.2	1	100	40	80.8	0	0	115.4	0.53
8-Jan	100	Mid	0	1	100	40	80.5	0	0	115	0.51
2-Feb	125	End	0	1	100	44	87.8	0	0	125.4	0.58
22-Feb	End	End	0	1	100	30	–	–	–	–	–

*Field irrigation efficiency = 70 %.

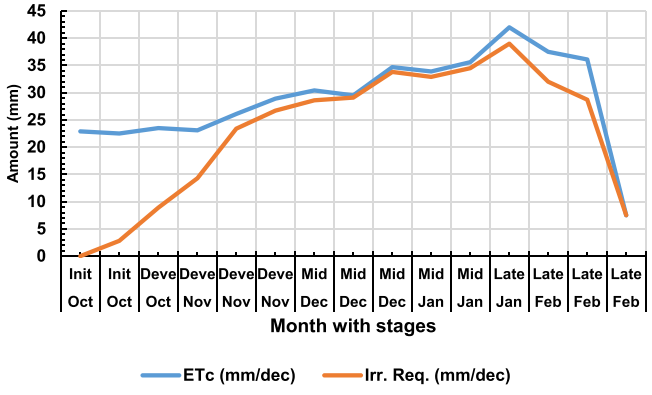


Fig. 8. Crop water requirement (ETc) and irrigation requirement for tomato. source: CropWat.

Kc drops from 1.1 to 0.8 from January 3 to February 3, reducing overall water consumption. While ETc remains high around 3.74 mm/day, increased rainfall somewhat reduces irrigation needs. The requirement drops from 39 mm during early January to just 7.5 mm at the start of February as the crop reaches maturity.

4.6. Irrigation schedule of tomato

The Table 6 & Fig. 9 tomato irrigation schedule follows a structured approach, ensuring soil moisture is replenished to field capacity at critical depletion levels. In the initial phase in October, moderate irrigation is required, with net irrigation increasing from 27.7 mm to 33.1 mm. Moreover, the soil moisture depletion reduces from 53 % to 31 %. With the development phase in November, water requirements increased to net irrigation of 55.8 mm, and soil depletion reduced to 36 %, indicating an increase in water consumption as the crop grows. In the mid-season stage, December and January show the period of peak water demand, with the highest applications of irrigation reaching 80.8 mm and 80.5 mm, at 40 % soil depletion. Gross irrigation again emphasizes water application in general, rising to a maximum of 115.4 mm. In the late stage, net irrigation stays high in February with 87.8 mm at 44 % soil depletion, stopping on 22 February due to crop maturity.

4.7. Crop water requirement (ETc) and Irrigation Water Requirement (IWR) of soybean

The crop water requirement analysis of soybeans in Chandrapur district, Maharashtra, is based on the given irrigation schedule (Table 7) & (Fig. 10). During the initial stage in June, the crop coefficient remains at 0.40, and ETc is relatively low during that time, the effective rainfall exceeds the water requirements of the crop, thereby making irrigation unnecessary. In the development stage, at the beginning of July, Kc increases to 0.53, and ETc goes to 26.6 mm. The effective rainfall during that time is 52.3 mm, which is more than enough to meet the water demand without needing any extra irrigation. The mid-season, from mid-July to mid-August, requires a great deal of water, and Kc reaches its maximum value of 1.10 and ETc attains 48.1 mm. On the contrary, the effective rainfall remains higher between 54.3 mm and 57.0 mm, which suffices to ensure that the crop receives adequate moisture without applying irrigation. From the late stage, August to September, the crop coefficient steadily declines, indicating lower water demand. During September, Kc dropped to 0.44, ETc reduced significantly to 1.8 mm. However, the declining effective rainfall also led to an irrigation requirement of 1.8 mm in the last phase of September. Therefore, Chandrapur experienced favorable rainfall conditions for soybeans for most of the crop growth cycle, making irrigation requirements very negligible except for a small requirement toward the end. As per the water balance considerations, the natural rainfall in the region adequately meets ETc

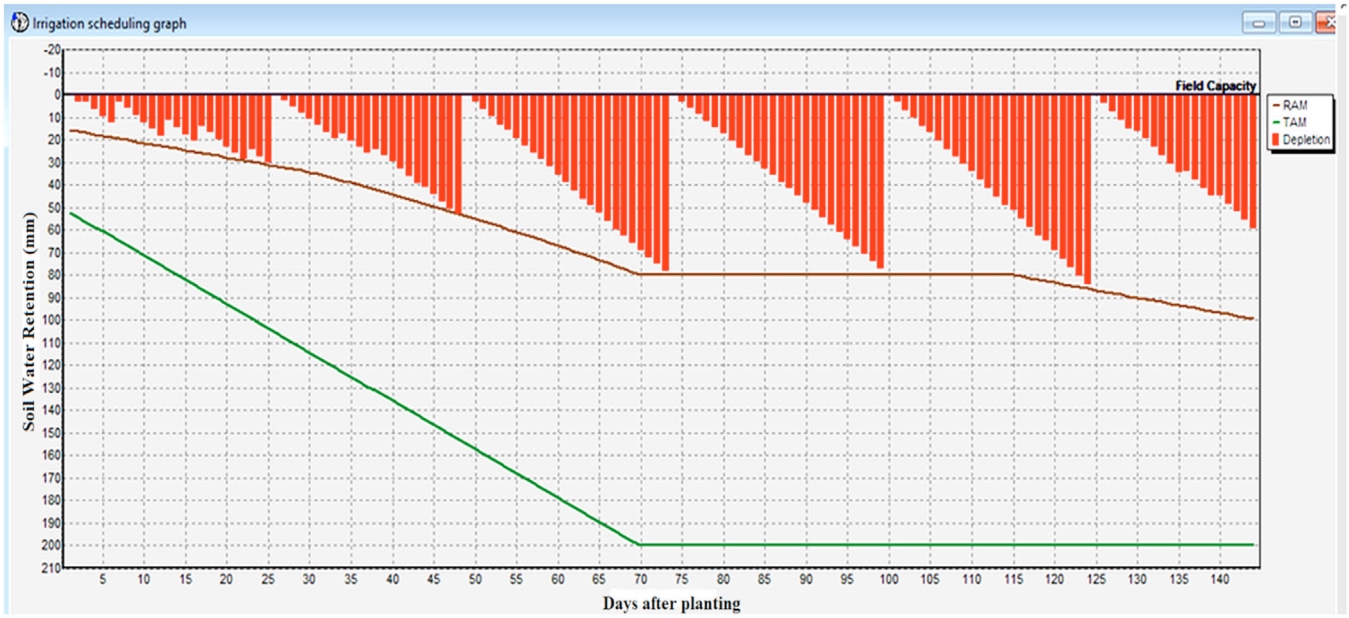


Fig. 9. Irrigation scheduling of the tomato crop, source: CropWat.
Here, RAM = Readily available moisture, TAM = Total available moisture [48].

Table 7
Month-wise crop water requirement and irrigation requirement of the soybean crop, source: CropWat.

Month	Decade	Stage	Kc (coeff)	ETc (mm/day)	ETc (mm/dec)	Eff rain (mm/dec)	Irr. Req. (mm/dec)
Jun	2	Init	0.4	2.9	5.8	9.3	5.8
Jun	3	Init	0.4	2.5	25	49.5	0
Jul	1	Deve	0.53	2.66	26.6	52.3	0
Jul	2	Mid	0.97	3.83	38.3	57	0
Jul	3	Mid	1.1	4.37	48.1	55.9	0
Aug	1	Mid	1.1	4.52	45.2	54.6	0
Aug	2	Mid	1.1	4.44	44.4	54.3	0
Aug	3	Late	1.06	4.26	42.6	52.8	0
Sep	1	Late	0.69	2.74	27.4	52.6	0
Sep	2	Late	0.44	1.76	1.8	5.2	1.8
		Total			309.3	443.2	7.6

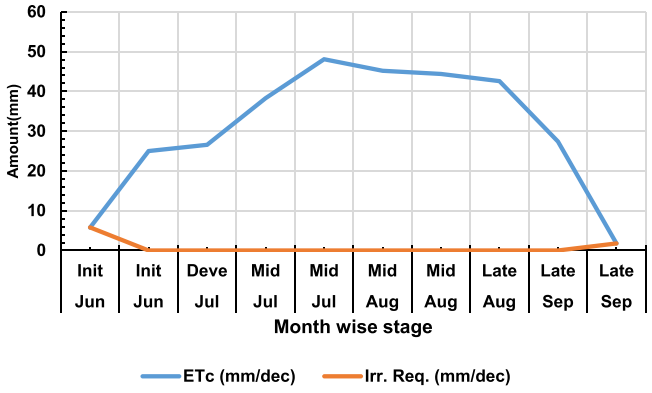


Fig. 10. Crop water requirement (ETc) and irrigation requirement of the soybean crop. source: CropWat.

or water requirement for soil moisture and good growth and yield of soybean without much dependence on artificial irrigation.

4.8. Irrigation schedule of soybean

The irrigation schedule of the soybean crop has shown that rainfall plays a vital role in meeting the water requirement. A total gross irrigation of 50.4 mm was applied, of which 35.3 mm was net irrigation with no losses and 100 % efficiency in irrigation scheduling. Total crop water use is 307.5 mm, which is equivalent to the crop's potential water requirement. Therefore, adequate water was available throughout the growing period. Rainfall contributed 943.7 mm, of which the crop effectively used 395.4 mm, and 548.3 mm was lost. The high rainfall resulted in a negative actual irrigation requirement of −87.9 mm, indicating that there was more than enough water received by the crop and therefore no additional irrigation needed. The efficiency of rainfall utilization was found to be 41.9 %, thus indicating that the crop did not effectively use a significant portion of the rainfall. Irrigation was applied to the initial stage on June 19, with 35.3 mm of net irrigation required due to 55 % of available soil moisture being depleted. However, by the end of the crop cycle on September 11, there was no requirement for additional irrigation (only a slight depletion of 5 %), and this was negligible. The irrigation schedule was highly efficient, and water demand was met by rainfall. Only a minimal quantity of irrigation was required subsequently in the early growth period. Careful management of water ensures a fruitful growth of soybeans (Table 8) & (Fig. 11).

In the present study, the crop water requirement (CWR) for tomatoes was estimated at 434.3 mm, with a total irrigation requirement of 342 mm. Additionally, the net irrigation requirement was found to be 365.7 mm, and the gross irrigation demand was 522.5 mm at 70 % field efficiency. These results are consistent with previous research conducted

Table 8
Stage-wise irrigation schedule of soybean crop, source: CropWat.

Date	Day	Stage	Rain (mm)	Ks (fract.)	Eta (%)	Depl (%)	Net Irr (mm)	Deficit (mm)	Loss (mm)	Gr. Irr (mm)	Flow (l/s/ha)
19-Jun	1	Init	0	1	100	55	35.3	0	0	50.4	5.84
11-Sep	End	End	0	1	100	5	–	–	–	–	–

*Field irrigation efficiency = 70 %.

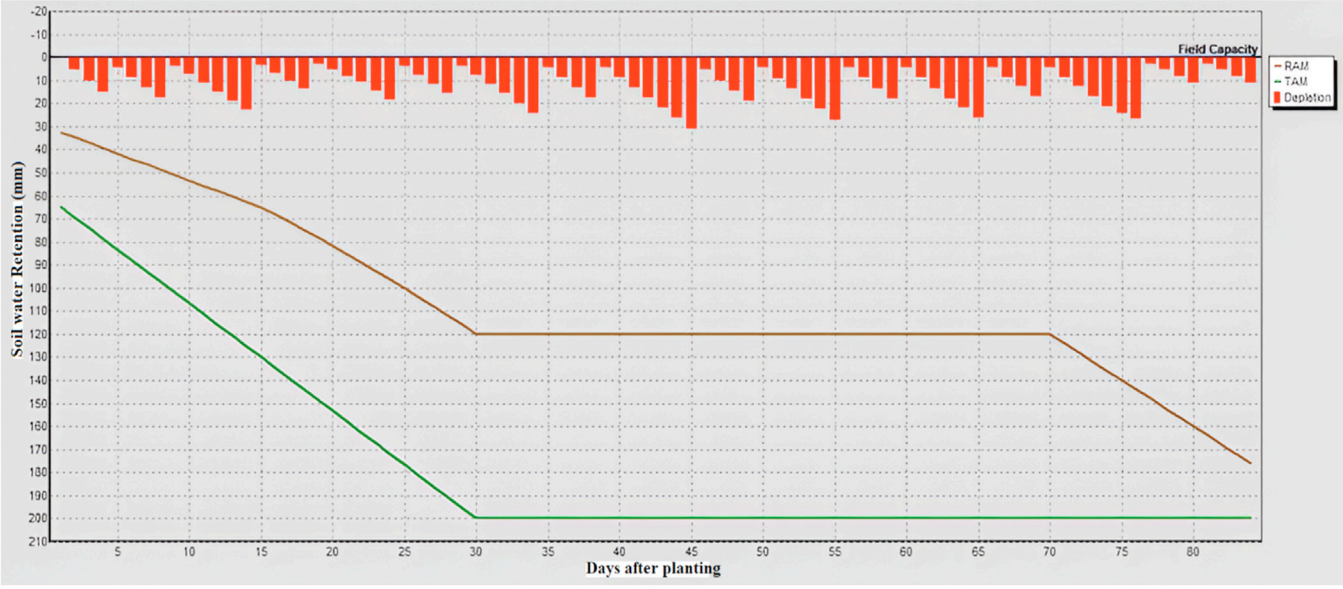


Fig. 11. Irrigation scheduling of soybean crop, source: CropWat.
Here, RAM = Readily available moisture, TAM = Total available moisture [48].

in similar agro-climatic regions. For instance, Ibrahim (2023) [49] estimated a CWR of 678.2 mm for tomato in Afgoye, Somalia, which is slightly higher due to regional climatic variations; however, the net irrigation (392.9 mm) and gross irrigation requirement (561.3 mm) values closely align with our findings. Similarly, Muroyiwa et al. (2022) [50] reported a gross irrigation requirement of 555.3 mm using the AquaCrop model for tomato in Harare, Zimbabwe. In India, Rana Pratap and Ch (2024) [51] estimated a gross irrigation requirement of 544.9 mm in Madakasira, Anantapur district of Andhra Pradesh for tomatoes, using the Cropwat model. Additionally, Ewaid et al. (2019) [19] reported a total irrigation requirement of 345.6 mm for tomato cultivation in southern Iraq. Another study by Shrikant et al. (2023), the result closely aligns with our findings for tomatoes CWR in Raichur region (375.64 mm) [32], demonstrating consistency and reinforcing the credibility of our analysis. For soybeans, the crop water requirement was calculated at 309.3 mm, the gross irrigation requirement is 50.4 mm, and the net irrigation requirement is 35.3 mm, which are in good agreement with findings from other regions. Gaddikeri et al. (2024) [52] reported soybean CWRs ranging from 303 mm to 380 mm across various agro-climatic zones of Madhya Pradesh using the CropWat model. Kumar et al. (2022) [53] found a CWR of 270.9 mm and net irrigation requirement of 10 mm in Sangareddy district, Telangana, indicating a result that is very similar to our study because of sufficient rainfall during the growing period. Moreover, Singh et al. (2014) [54] estimated a slightly higher soybean CWR of 401.6 mm under vertisols soil conditions in Madhya Pradesh, likely due to longer crop duration and different soil-water dynamics. According to Phad et al. (2024), who found the crop water requirement of soybeans range from 419.60 mm in Aurangabad to 457.81 mm in Parbhani for the Marathwada region [35], this higher variation from our result may be attributed to differences in regional climatic conditions, soil characteristics and crop management practices but still reliable to support our study, because the crop water requirement (CWR) of soybean usually varies between 300 mm to 800 mm [65]. These comparative analyses support the reliability of our findings and highlight the applicability of the CropWat 8.0 model for irrigation planning across diverse agroclimatic zones.

5. Implications of this study for the agricultural environment

This study has a significant implication for agricultural environment principles in irrigation planning [55–57]. Advanced water use in agriculture improves the yields of crops and reduces the stress on local aquifers, which minimizes the adverse effects from over-extraction of water [58]. Inefficient irrigation inevitably leads to soil degradation, water scarcity, and ecological imbalance in semi-arid locations like Chandrapur. This study promoted water-use efficiency by optimizing irrigation scheduling with FAO CropWat 8.0 and supporting balanced hydrological cycles. Additionally, the study is pertinent to the field of water ecology as it addresses irrigation-induced ecological ramifications on soil and water quality. Excess irrigation is practically always associated with nutrient leaching, soil salinization, and groundwater pollution, all of which adversely affect the aquatic ecosystems, microbial diversity, and plant health [59–61]. On the other hand, a lack of water may adversely influence geographical vegetation patterns, evapotranspiration cycles, and local microclimates in agriculture. This work promotes ecological sustainability in agricultural environments by assessing the water requirements for tomatoes and soybeans to achieve adequate irrigation scheduling.

6. Awareness of farmers

According to Tables 6 & 8, irrigation scheduling may be a great advantage to farmers in choosing the correct amount and frequency of

irrigation water. Farmers should have gone through the scientific awareness campaign conducted by the local authorities that will educate them regarding the suitable crops for various conditions in order to make optimal use of available water [62,63]. Thus, it is possible to save a significant quantity of irrigation water through matching the availability of actual water for irrigation with crop water demand.

7. Conclusion

The crop water requirements and irrigation scheduling of tomatoes and soybeans are very essential for enhancing yield, water efficiency, and general crop health. The water demands and optimum irrigation scheduling for tomatoes and soybeans are highlighted in the study using FAO CropWat 8.0, considering their water needs and growth stages, as well as regional climatic conditions. The work was carried out in Chandrapur, Maharashtra, in a semiarid climatic zone with variable rainfall patterns, where sustainable agriculture is dependent upon efficient water management. It was exercised with calculations of water requirements based on ET_o and monitoring of soil moisture to ascertain precise water needs and irrigation scheduling. Tomatoes need a high volume of water, as they demand a regular supply of moisture, especially during flowering and the fruiting period, with (434.3 mm/dec) of water requirements and irrigation required (342mm/dec) in every season. Soybean is an indigenous drought-tolerant crop requiring water of about 309.3mm/dec and 7.6mm/dec for irrigation. Irrigation scheduling is mainly at the flowering and pod-filling stages for soybean crops. On the other hand, the CropWat model projected a water need of 35.3 mm net irrigation and 50.4 mm gross irrigation at 70 % field efficiency for thrive production for Soybean, and for tomato, a water need of 365.7 mm net irrigation and 522.5 mm gross irrigation at 70 % field efficiency for best production. Arranging irrigation, suitably in relation to ETo, soil moisture monitoring, and climatic conditions, allows for sustainable water use in this region. Using modern scientific tools such as CropWat 8.0, the water requirement of the crops can be easily determined with greater accuracy. Therefore, this research is very fruitful to researchers, policymakers, and farmers.

8. Future research

While this study successfully demonstrates the application of the FAO CropWat 8.0 model to determine crop water requirements and irrigation scheduling for tomato and soybean in Chandrapur district. Several avenues remain open for further research. Therefore, future studies could integrate high-resolution spatial data obtained from remote sensing and GIS to develop location-specific irrigation strategies. The incorporation of real-time soil moisture sensors, an automated irrigation system, and AI-driven decision support tools can further enhance the precision of water use. Moreover, assessing the impact of climate change on crop evapotranspiration and irrigation demand through long-term climatic modelling will be crucial to ensure resilience in semi-arid agricultural practices. In addition, region-specific local Kc values would enhance the model results more in future research. On another context, Engaging local farming communities in participatory research and capacity-building initiatives would facilitate the translation of model-based insights into practical and scalable interventions for sustainable water resource management.

Declaration of interest

Declare that there are no conflicts of interest that could have influenced the research presented in this study.

• We have no financial, commercial, personal, or institutional relationships that could be perceived as influencing the content of this work.

• No funding sources, grants, or sponsorships have created any bias in conducting, analyzing, or presenting this research.

• All necessary ethical considerations have been followed, and any relevant disclosures have been made transparently.

CRediT authorship contribution statement

Sourajit Mal: Writing – original draft, Writing – review & editing, Resources, Methodology, Visualization. **Soumita Sen:** Writing – original draft, Writing – review & editing, Resources, Methodology, Visualization.

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